Saachi Kokate
T13-50
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SEPM Assignment 2

Aim: To understand DevOps: Principles, Practices, and DevOps Engineer Role and Responsibility

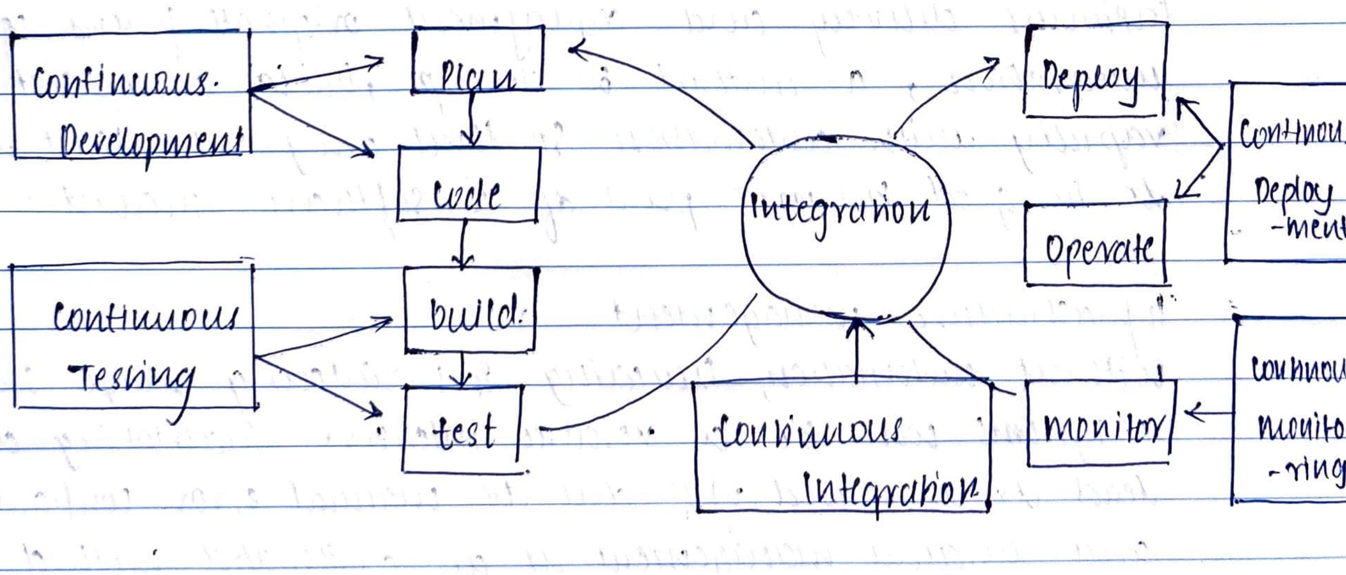
What is DevOps?

DevOps is a collaborative approach where teams work together to build and deliver secure software efficiently. It combines software development (dev) and operations (Ops) to decide how to accelerate delivery through automation, collaboration, fast feedback, and iterative improvement. Built on Agile methodology, DevOps creates a culture of accountability, collaboration, and shared responsibilities for business outcomes.

Core Principles:

1. ~~both~~ Develop and test in production-like environments.
2. Deploy builds frequently
3. Continuously validate operational quality

DevOps Practices:



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Continuous Development
This is the phase that involves planning and coding, versioning and managing build of the software application's functionality.
ex:- git, github, maven, etc.

Continuous Testing
Continuous testing is, executing automated tests, continuously and repeatedly against the code base and the various deployment environments. It is a software testing methodology which focuses on achieving continuous quality & improvement.
ex:- Bamboo, appium.

Continuous Integration.
Continuous integration refers to the build and unit testing stages of the software release process. Every revision that is committed triggers an automated build and test.
ex:- Jenkins, Travis CI, circleci.

Continuous Delivery & Deployment
Continuous delivery and deployment originate from continuous integration, a method to develop, build and test new code rapidly with automation so that only code that is known to be good becomes part of a software product.

Infrastructure management
Without automation, building & maintaining large-scale modern IT systems can be a resource-intensive undertaking and can lead to increased risk due to manual error. configuration and resource management is an automated method for

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maintaining computer systems and software in a known, consist-ent state.

Configuration management.
Infrastructure as code is the practise of describing all software runtime environment and networking settings and parameters in simple textual format, that can be stored in your version control system (VCS) and versioned on request. These text files are called manifest and are used by DevOps tools to automatically provision & configure build serves, testing & production environment.

Microservice Architecture
Docker is a tool designed to make it easier to create, deploy, and run application by using containers. Containers allows a developer to package up an application with all of the parts it needs, such as libraries and other dependencies and deploy it as one package. By doing so, thanks to the container the developer can rest assured that the application will run on any other Linux machine regardless of any customized settings that machine might have.
ex:- Nagios, splunk, etc.

Cloud Based DevOps.
DevOps automation is becoming cloud centric. Most public and private cloud computing provider support DevOps systematically on their platform, including continuous integration and continuous development tools.
ex:- amazon web services, amazon lambda, ~~gog~~ google cloud, etc

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Dev Ops Engineer Roles:

A DevOps engineer manages a company's IT infrastructure, bridging development and operation. Key responsibilities include:-

Technical Responsibilities:

- Implement development, testing and automation tools.
- Set up infrastructure and tools.
- Code review and responsibilities.
- Bug fixing and trouble shooting
- Build and maintain CI/CD pipelines.
- Security implementation and monitoring

Management Responsibilities:

- Understand customer requirements and KPIs.
- Plan team structure and activities.
- Manage stakeholders
- Define development and operational processes
- Co-ordinate team communication.
- Monitor customer experience
- Provide periodic progress reports
- Mentor team members.